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LabGym Triage Action Plan

Dataset Target: sorted_mouse_behavior_examples | Date:

07/14/2026

Last Updated: 07/14/2026

Overall Model Diagnostics Summary:

Accuracy: **83.6%** | Macro F1-Score: **80.9%**

Confusion Matrix Overview

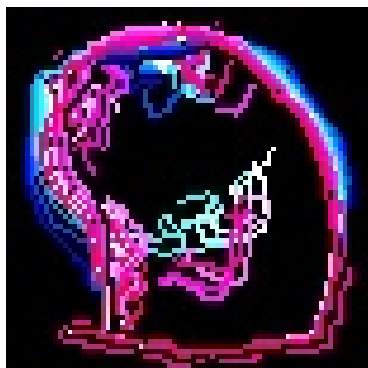
Cells with gold borders highlight systematic class confusions addressed in diagnostics.

True \ Pred	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
1. behind the wheel	164	0	0	0	0	1	0	0	0	5	1	1	0	0	1	0	0	0	2	2
2. body grooming fv	0	120	6	8	0	0	121	4	99	1	0	0	3	5	0	0	0	0	0	0
3. body grooming sv	0	4	141	5	2	0	8	39	45	2	0	0	1	1	0	1	0	2	3	0
4. chewing fv	0	1	2	96	6	0	8	1	6	1	0	0	0	2	0	0	0	0	0	0
5. chewing sv	0	0	2	0	172	0	0	2	0	19	0	0	0	0	2	0	0	0	3	0
6. coming down	0	0	0	0	0	367	0	0	0	0	3	11	2	0	1	1	1	0	0	1
7. face grooming fv	0	3	0	0	0	0	144	1	13	0	0	0	0	2	0	1	0	0	0	0
8. face grooming sv	0	1	56	30	3	0	1	78	1	3	0	0	0	2	0	0	0	0	1	0
9. foraging fv	0	2	1	4	1	0	1	0	347	5	0	0	0	0	1	1	3	2	0	0
10. foraging sv	0	0	0	0	3	0	0	1	2	492	0	0	0	0	0	1	0	0	1	0
11. rearing up	0	0	0	0	0	2	2	0	1	0	407	7	2	2	3	2	0	2	0	0
12. resting on the wheel	1	0	0	0	0	2	0	0	0	0	1	308	10	1	1	0	0	1	1	5
13. running on the wheel	0	0	0	0	0	0	0	0	0	0	0	4	419	0	0	0	0	0	0	1
14. sniffing fv	0	1	1	1	0	1	5	2	1	1	2	0	2	85	5	5	1	3	0	0
15. sniffing sv	0	0	0	0	0	2	1	0	0	13	4	30	3	1	204	8	0	1	1	1
16. standing	0	0	0	0	0	9	0	0	0	1	6	44	11	0	2	305	0	3	0	0
17. turning front	0	0	0	0	0	0	0	0	2	6	0	1	0	1	0	0	65	3	0	1
18. turning side	0	0	2	0	1	2	0	0	2	15	1	2	8	1	3	2	4	198	11	5
19. unknown bv	0	1	3	0	1	0	0	0	3	5	0	0	0	0	0	1	0	1	377	0
20. walking	0	0	0	0	0	0	0	0	0	5	2	0	4	0	0	0	1	4	0	224

Systematic Major Confusions (Auto-Centroids)

Below are the systematic major confusions (where errors exceed 10% of class support) detected in this test run. For each confusion cell, the mathematical centroid has been calculated using Cosine Similarity to capture

the structural cluster orientation of behavior cells in the high-dimensional feature space.



body grooming fv -> face grooming fv

Detected systematic major confusion of body grooming fv classified as face grooming fv.

Error metrics: **121 errors** (33.0% error proportion).

Centroid video: 102_body grooming fv_false.avi

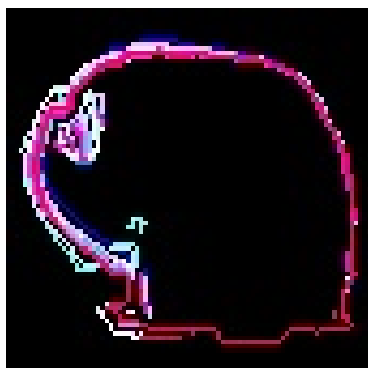


body grooming fv -> foraging fv

Detected systematic major confusion of body grooming fv classified as foraging fv.

Error metrics: **99 errors** (27.0% error proportion).

Centroid video: 139_body grooming fv_false.avi



body grooming sv -> face grooming sv

Detected systematic major confusion of body grooming sv classified as face grooming sv.

Error metrics: **39 errors** (15.4% error proportion).

Centroid video: 191_body grooming sv_false.avi



body grooming sv -> foraging fv

Detected systematic major confusion of body grooming sv classified as foraging fv.

Error metrics: **45 errors** (17.7% error proportion).

Centroid video: 219_body grooming sv_false.avi

**face grooming sv -> body grooming sv**

Detected systematic major confusion of face grooming sv classified as body grooming sv.

Error metrics: **56 errors** (31.8% error proportion).

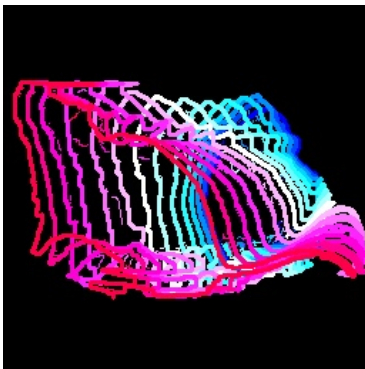
Centroid video: 72_face grooming sv_false.avi

**face grooming sv -> chewing fv**

Detected systematic major confusion of face grooming sv classified as chewing fv.

Error metrics: **30 errors** (17.0% error proportion).

Centroid video: 55_face grooming sv_false.avi

**sniffing sv -> resting on the wheel**

Detected systematic major confusion of sniffing sv classified as resting on the wheel.

Error metrics: **30 errors** (11.2% error proportion).

Centroid video: 17_sniffing on the wheel_std50.avi

Hypothesis 1: Redundant Behaviors

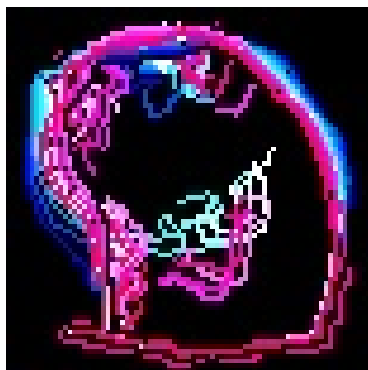
These behaviors look identical to the software. No further action is required.

None selected.

Hypothesis 2: Emergent Behaviors

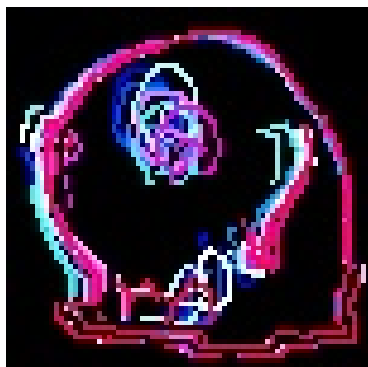
The software is confusing these behaviors because there is likely a third, completely different behavior happening that it hasn't been taught yet.

ACTION REQUIRED: Go to the video folders and manually move the mixed-up videos into a new behavior folder.

**body grooming fv -> face grooming fv**

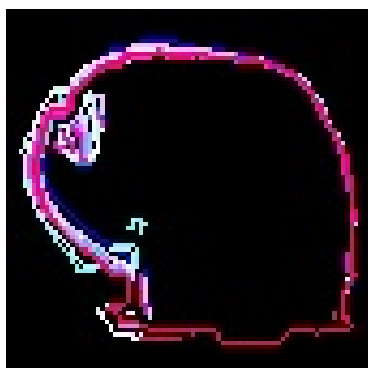
Confusion suggests emergent behavior subclass may be present.

Reference centroid file: 102_body grooming fv_false.avi

**body grooming fv -> foraging fv**

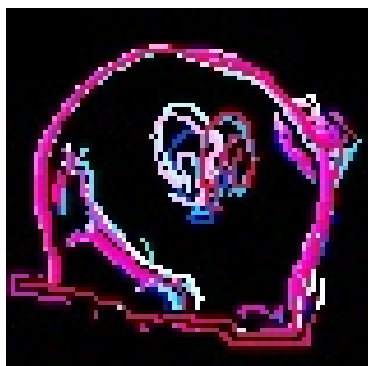
Confusion suggests emergent behavior subclass may be present.

Reference centroid file: 139_body grooming fv_false.avi

**body grooming sv -> face grooming sv**

Confusion suggests emergent behavior subclass may be present.

Reference centroid file: 191_body grooming sv_false.avi

**body grooming sv -> foraging fv**

Confusion suggests emergent behavior subclass may be present.

Reference centroid file: 219_body grooming sv_false.avi

**face grooming sv -> body grooming sv**

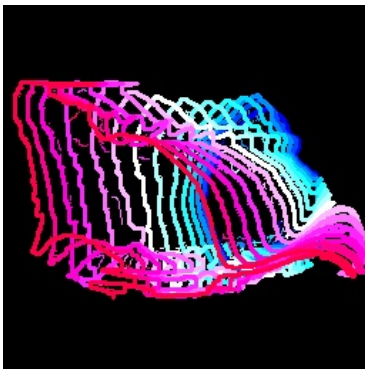
Confusion suggests emergent behavior subclass may be present.

Reference centroid file: 72_face grooming sv_false.avi

**face grooming sv -> chewing fv**

Confusion suggests emergent behavior subclass may be present.

Reference centroid file: 55_face grooming sv_false.avi

**sniffing sv -> resting on the wheel**

Confusion suggests emergent behavior subclass may be present.

Reference centroid file: 17_sniffing on the wheel_std50.avi

Hypothesis 3: Poor Generalization

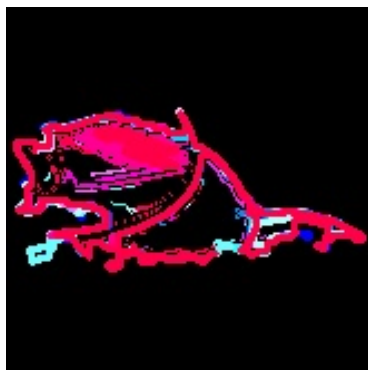
The software is failing here because it hasn't seen enough different examples of this behavior.

ACTION REQUIRED: Add more videos of these specific behaviors in different environments or angles.

None selected.

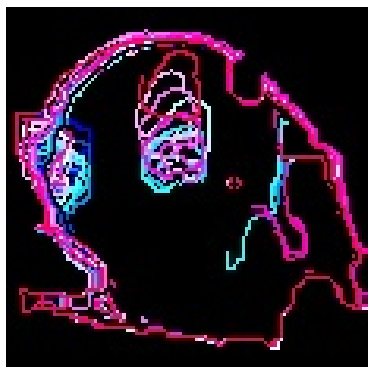
Baseline References (True Positives)

Below are representative examples of correctly classified behaviors. Use these as a visual baseline for what the model currently considers an 'ideal' representation of the class.

***behind the wheel (True Positive)***

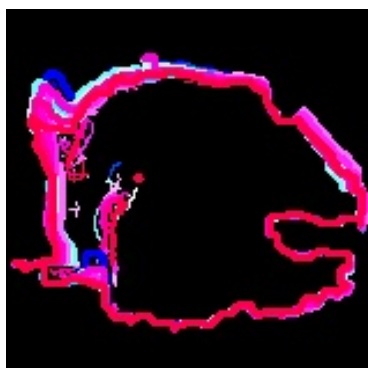
Representative baseline centroid frame showing ideal class representation.

Reference centroid file: 91_behind the wheel_std50.avi

***body grooming fv (True Positive)***

Representative baseline centroid frame showing ideal class representation.

Reference centroid file: 41_body grooming fv_std50.avi

***body grooming sv (True Positive)***

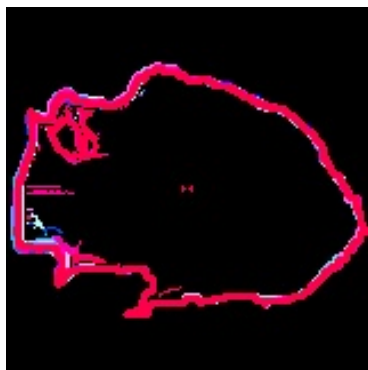
Representative baseline centroid frame showing ideal class representation.

Reference centroid file: 55_body grooming sv_std50.avi

***chewing fv (True Positive)***

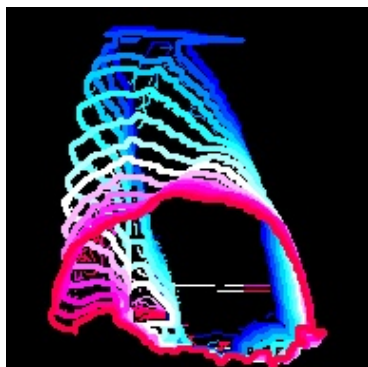
Representative baseline centroid frame showing ideal class representation.

Reference centroid file: 16_chewing fv_std50.avi

***chewing sv (True Positive)***

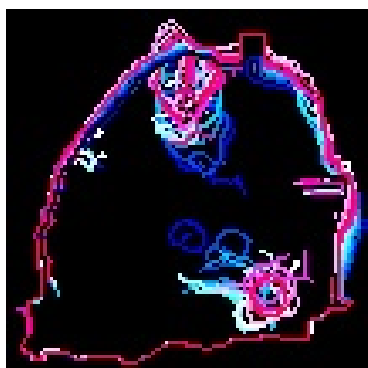
Representative baseline centroid frame showing ideal class representation.

Reference centroid file: 91_chewing sv_std50.avi

***coming down (True Positive)***

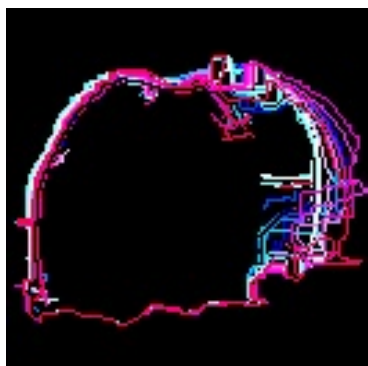
Representative baseline centroid frame showing ideal class representation.

Reference centroid file: 62_coming down sv_std50.avi

***face grooming fv (True Positive)***

Representative baseline centroid frame showing ideal class representation.

Reference centroid file: 103_face grooming fv_std50.avi

***face grooming sv (True Positive)***

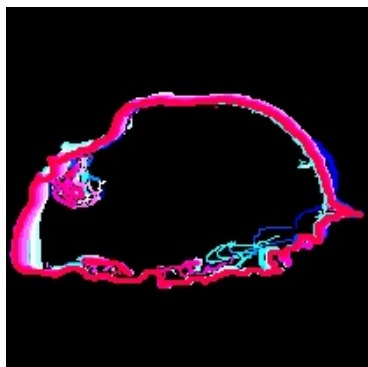
Representative baseline centroid frame showing ideal class representation.

Reference centroid file: 20_face grooming sv_std50.avi

***foraging fv (True Positive)***

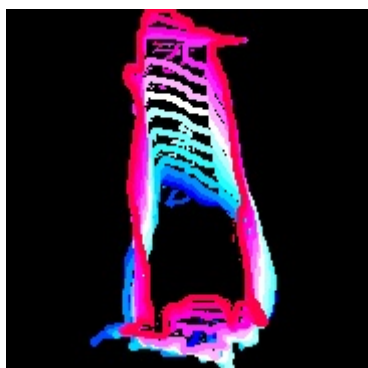
Representative baseline centroid frame showing ideal class representation.

Reference centroid file: 82_foraging fv_std50.avi

***foraging sv (True Positive)***

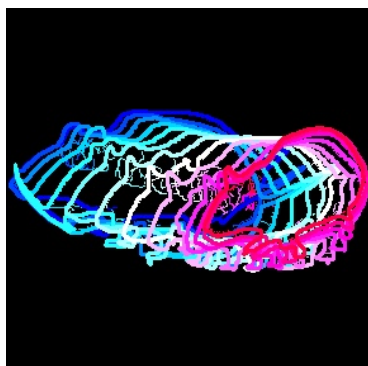
Representative baseline centroid frame showing ideal class representation.

Reference centroid file: 80_foraging sv_std50.avi

***rearing up (True Positive)***

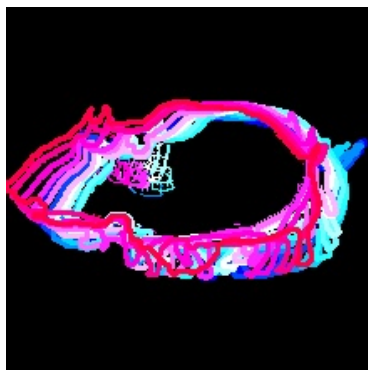
Representative baseline centroid frame showing ideal class representation.

Reference centroid file: 105_rearing up sv_std50.avi

***resting on the wheel (True Positive)***

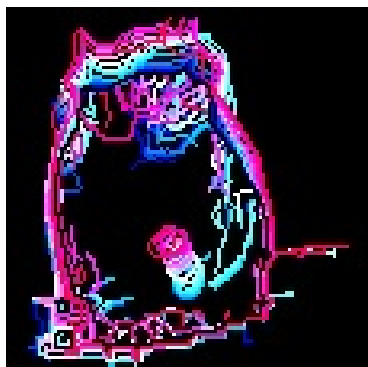
Representative baseline centroid frame showing ideal class representation.

Reference centroid file: 96_resting on the wheel_std50.avi

**running on the wheel (True Positive)**

Representative baseline centroid frame showing ideal class representation.

Reference centroid file: 292_running on the wheel_std50.avi

**sniffing fv (True Positive)**

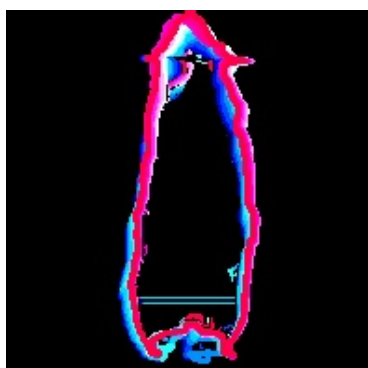
Representative baseline centroid frame showing ideal class representation.

Reference centroid file: 86_sniffing fv_std50.avi

**sniffing sv (True Positive)**

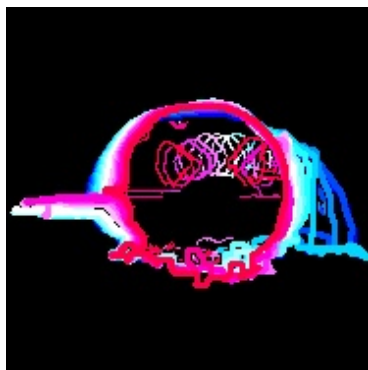
Representative baseline centroid frame showing ideal class representation.

Reference centroid file: 67_sniffing sv_std50.avi

**standing (True Positive)**

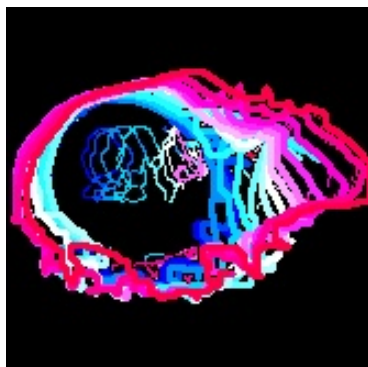
Representative baseline centroid frame showing ideal class representation.

Reference centroid file: 78_standing bv_std50.avi

**turning front (True Positive)**

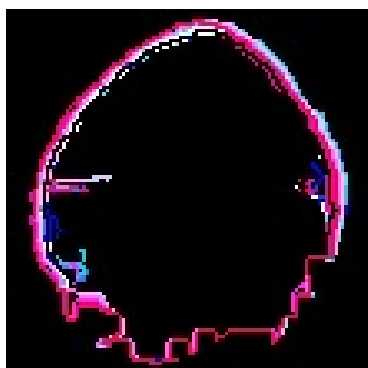
Representative baseline centroid frame showing ideal class representation.

Reference centroid file: 24_turning front_std50.avi

**turning side (True Positive)**

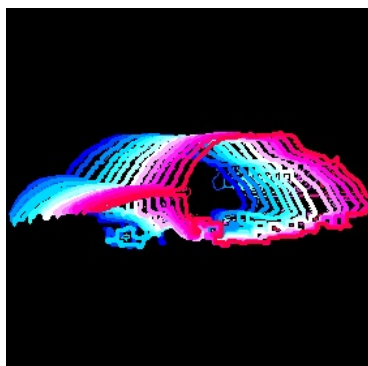
Representative baseline centroid frame showing ideal class representation.

Reference centroid file: 130_turning side_std50.avi

**unknown bv (True Positive)**

Representative baseline centroid frame showing ideal class representation.

Reference centroid file: 179_unknown bv_std50.avi

**walking (True Positive)**

Representative baseline centroid frame showing ideal class representation.

Reference centroid file: 70_walking_std50.avi